

CST3340

Business Intelligence

Coursework 2

Name

Module Leader: Stephen Agada

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1. Introduction

This project analyzes the QS World University Rankings dataset (2017–2027), one of the most widely recognized global university ranking systems, providing comparative measures of institutional performance across teaching, research, employability, and international reputation. Separate annual datasets from Kaggle were merged into a single master dataset of 13,910 records to enable longitudinal analysis across an eleven-year period. This analysis aims to identify long-term trends, regional differences, and institutional performance patterns, offering business intelligence relevant to prospective students choosing institutions, university leaders benchmarking against competitors, and policymakers assessing national higher education competitiveness.

The dataset includes categorical variables like university, country, continent, and year, along with numerical variables such as overall score, numeric rank, percentile rank, faculty-student ratio, academic reputation, employer reputation, citations per faculty, and sustainability score. QS's methodology has changed over the years. Academic reputation, employer reputation, citations, and faculty-student ratio have only been consistently available since 2023. Sustainability scores are entirely missing for 2023 and 2025.

Several cleaning steps were needed before importing the data into Tableau. We standardized column names across annual files, normalized country names, and corrected inconsistent spellings and formats of university names over the years. This ensured each institution tracked as a single, consistent entry for long-term analysis. Universities assigned rank ranges, such as 601 to 610, were adjusted to midpoints like 605.5, while the open-ended 1401+ category received a minimum value of 1401. Additionally, the number of ranked universities rose from 933 in 2017 to 1,504 by 2027. This increase makes direct rank comparisons over the years misleading. To address this, a Percentile Rank calculated field was created by dividing each university's numeric rank by the total number of ranked universities that year, allowing for fair comparisons over time. Data cleaning and integration were done in Python using Google Colab, while visual analytics were created in Tableau.

2. Data Analysis and Visualisation (35 marks) (approx. 840-1050 words)

Ten visualizations were developed in Tableau, progressing from global and temporal overviews toward institution-level detail. Together, these identify patterns providing business intelligence for universities, policymakers, and prospective students.

Global Distribution of Top University Performance (2023-2027)

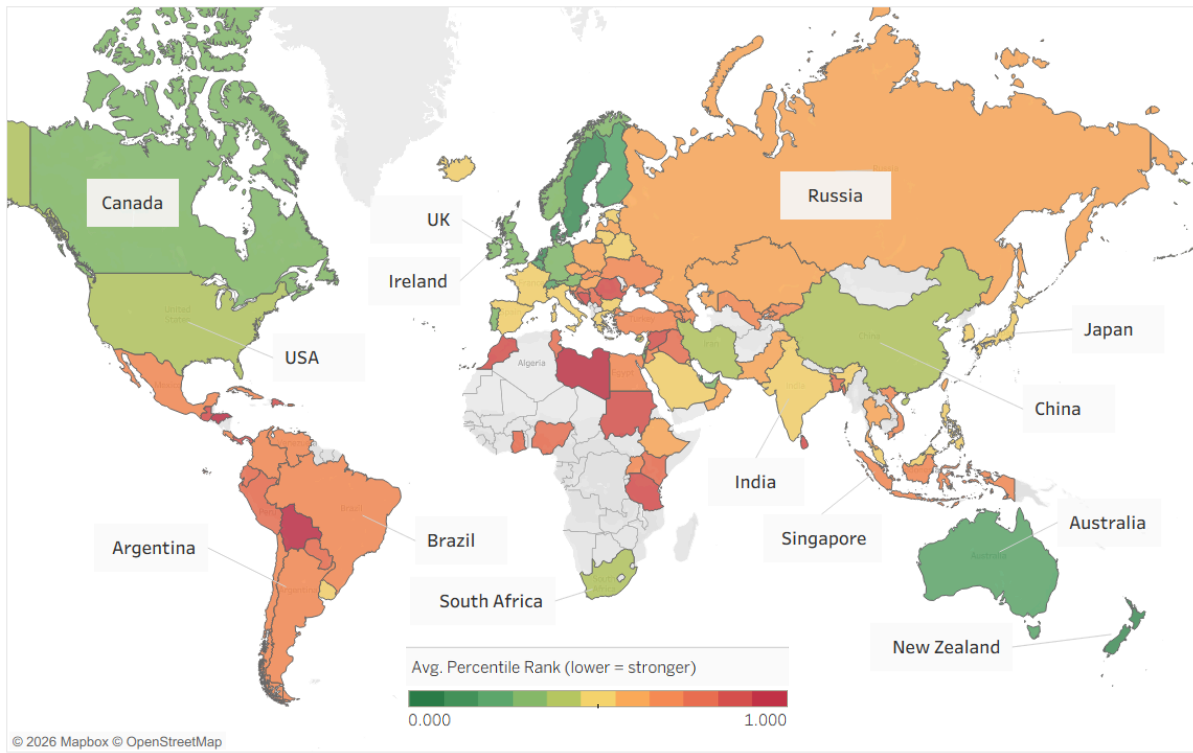


Figure 1. Filled Map: Global Distribution of Top University Performance (2023-2027)

Figure 1 shows each country's average Percentile Rank from 2023 to 2027, using Percentile Rank rather than raw ranking to allow fair comparison given the substantial growth in ranked universities over the period. Strong performance concentrates in Oceania, Northern and Western Europe, China, and North America. Meanwhile, the Caribbean, Northern and Eastern Africa, and Latin America show at least one ranked university but typically at low percentile, while most of Africa remains unrepresented, confirming global prestige remains concentrated despite the rankings' expansion.

Average Top University Performance by Region Over Time (2017-2027)

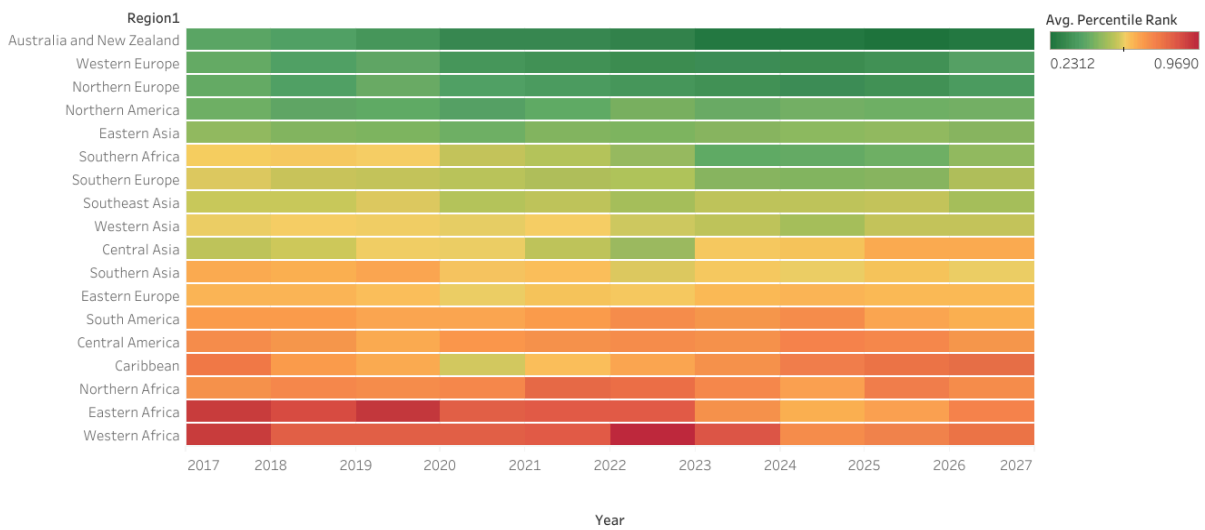


Figure 2. Heatmap: Average Top University Performance by Region Over Time (2017-2027)

Figure 2 breaks this down further, heatmapping Percentile Rank across all 18 QS sub-regions from 2017–2027. Western Europe and Northern America maintained consistently strong positions, while African sub-regions show minimal, volatile representation due to small ranked-university counts. Regional strength has remained unevenly distributed and changed only gradually over the eleven years.

Top 20 Countries by Avg. Percentile Rank (2023-2027)

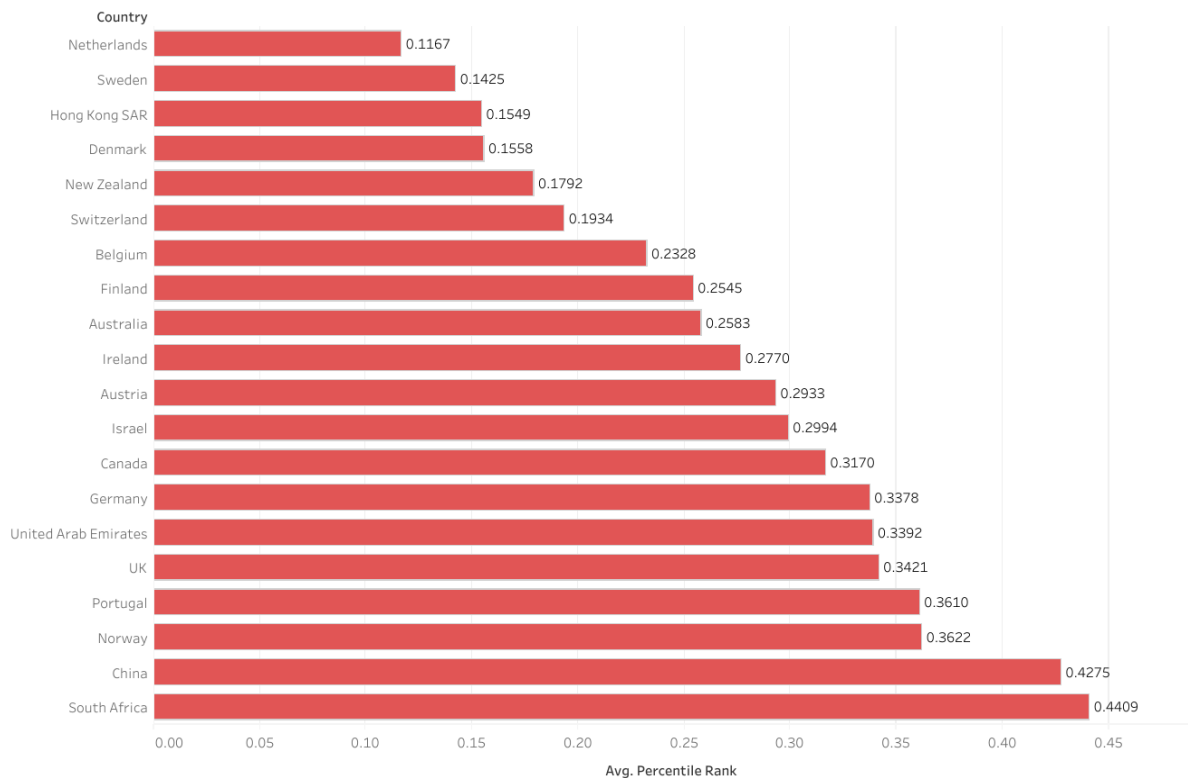


Figure 3. Bar Graph: Top 20 Countries by Average Percentile Rank (2023-2027)

Figure 3 narrows to 2023–2027, ranking the top 20 countries by average Percentile Rank (minimum five ranked universities, excluding small systems like Qatar). The Netherlands leads (0.1167), with Nordic and Western European countries dominating; the UK ranks lower than its reputation might suggest. Resource-concentrated nations, which are Singapore, Switzerland, Hong Kong, outperform much larger systems, indicating national scale is not a reliable predictor of higher-education strength.

Ranking Stability Distribution of Top 20 Countries (2023-2027)

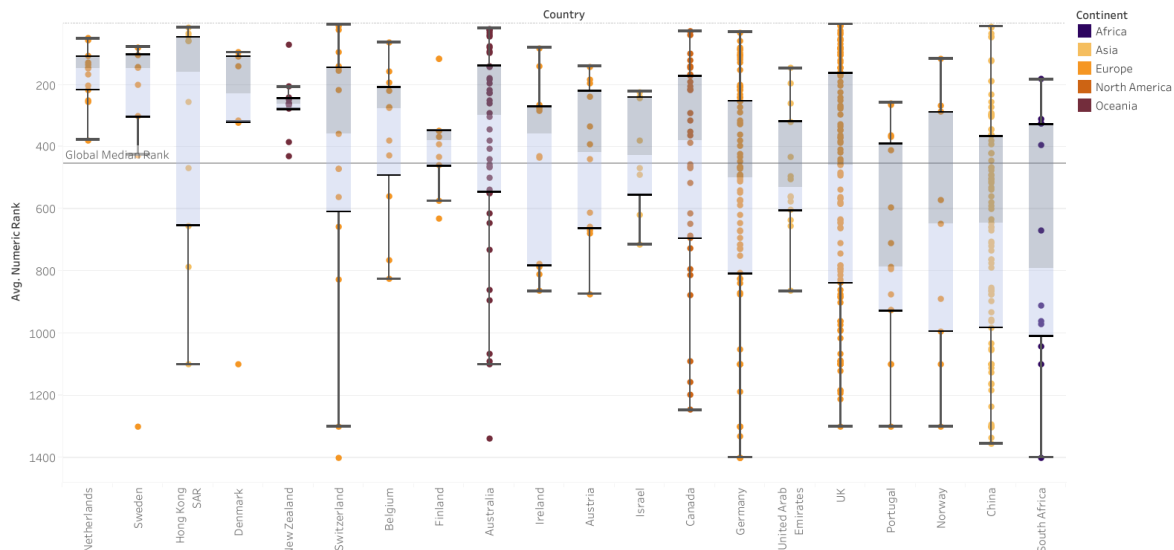


Figure 4. Box Plot: Ranking Stability Distribution of Top 20 Countries (2023-2027)

Figure 4, a box-and-whisker plot of the average Numeric Rank for these 20 countries, shows the Netherlands, Sweden, Denmark, New Zealand, and Finland sitting above the global median with small interquartile ranges (IQR), showing consistent, tightly clustered performance. Hong Kong and Switzerland, despite ranking highly, show much wider spread. The UK, Germany, and China show the largest IQRs, reflecting a mix of elite and lower-performing institutions. This distinguishes uniformly strong systems from those propped up by a handful of elite institutions.

Top 20 Country Strengths Across QS Metrics (2023-2027)

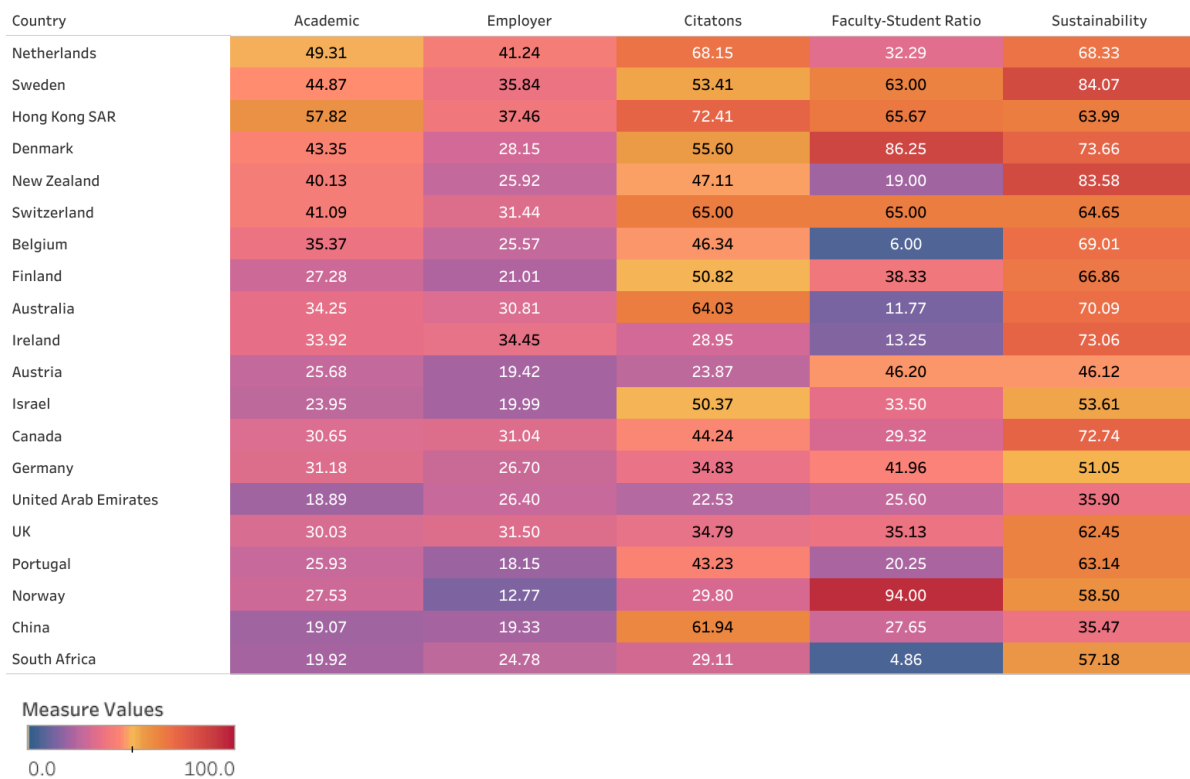


Figure 5. Highlight Table: Top 20 Country Strengths Across QS Metrics (2023-2027)

Figure 5 drills into the five metrics driving overall score: Academic Reputation, Employer Reputation, Citations, Faculty-Student Ratio, and Sustainability. No country dominates every metric. Hong Kong performs strongly across nearly all five, while South Africa and Belgium post very low Faculty-Student scores despite reasonable performance elsewhere, and Nordic countries lead on Sustainability. In all countries, Employer Reputation skews on the lower side while Sustainability skews high. This suggests countries pursue strategic specialization rather than broad dominance.

Average Performance Over Time Among English-Medium Countries (2017-2027)

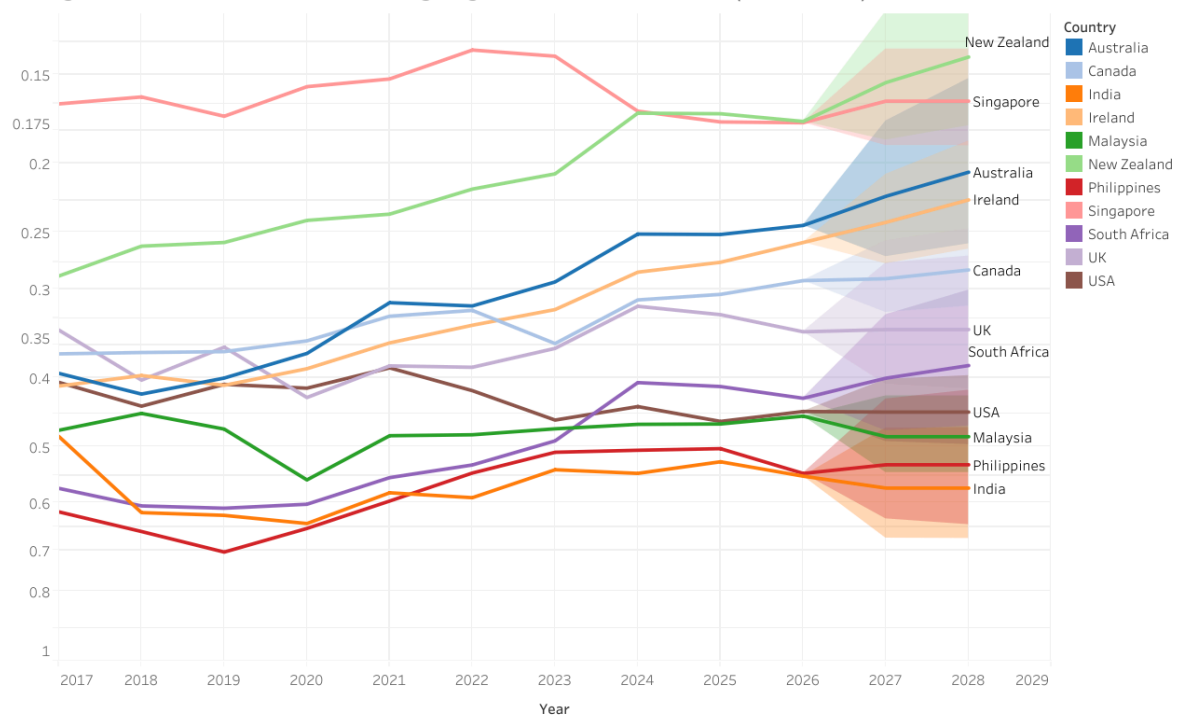


Figure 6. Line Graph: Average Performance Over Time Among English-Medium Countries (2017-2027)

Figure 6 tracks a selected set of eleven English-medium countries (2017–2027). The countries include the Core Anglosphere alongside select English-medium countries (Singapore, South Africa, Malaysia, Philippines, India) chosen for sustained QS participation. New Zealand shows the strongest improvement, finishing top; Singapore ranks second, ahead of Australia, Ireland, Canada, and the UK. The USA sits near the bottom, behind South Africa, likely reflecting an averaging effect, as Singapore's average reflects a few elite institutions while the USA's 184 universities span a much broader range. Malaysia, the Philippines, and India occupy the lower end, though forecasts suggest gradual improvement. Findings indicate that English-medium status alone does not determine competitiveness; system size and national investment matter more.

Top 100 University Concentration by Country (2027)

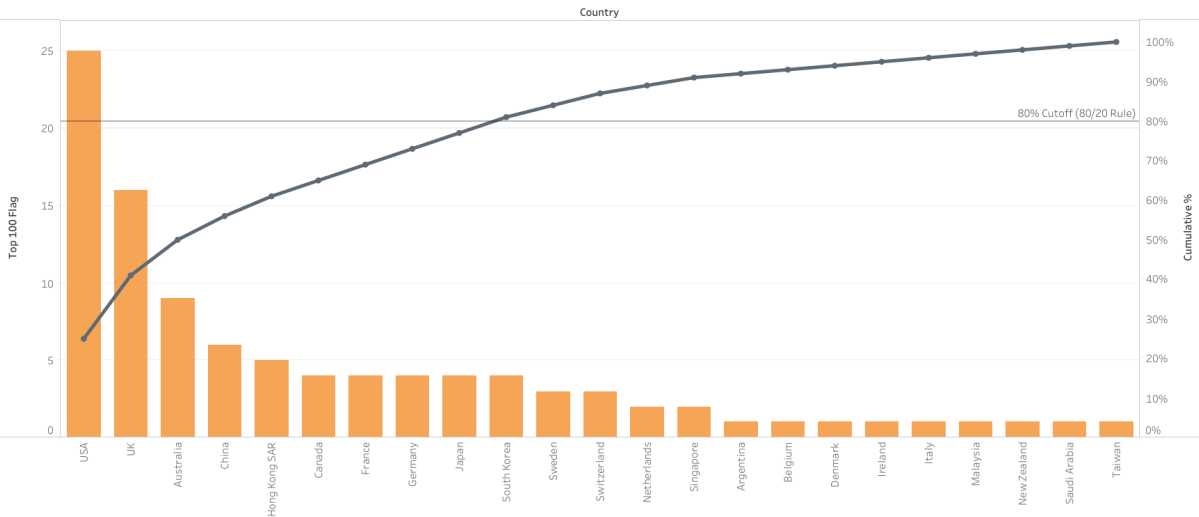


Figure 7. Pareto Chart: Top 100 University Concentration by Country (2027)

Figure 7 shows a Pareto Chart that highlights the countries where the world's Top 100 universities are located as of 2027. The chart only includes countries with at least one university in the Top 100, ranked by their number. Each bar represents the total number of universities from that country. An overlaid line tracks the total percentage of Top 100 universities, with a reference line indicating the 80% mark, which confirms the classic Pareto 80/20 principle. The results show that just nine countries, the USA, UK, Australia, China, Hong Kong, Canada, France, Germany, and Japan, together make up 80% of the world's Top 100 universities in 2027, revealing a strong concentration effect.

Tier Mobility in Global Higher Education (2017-2027)

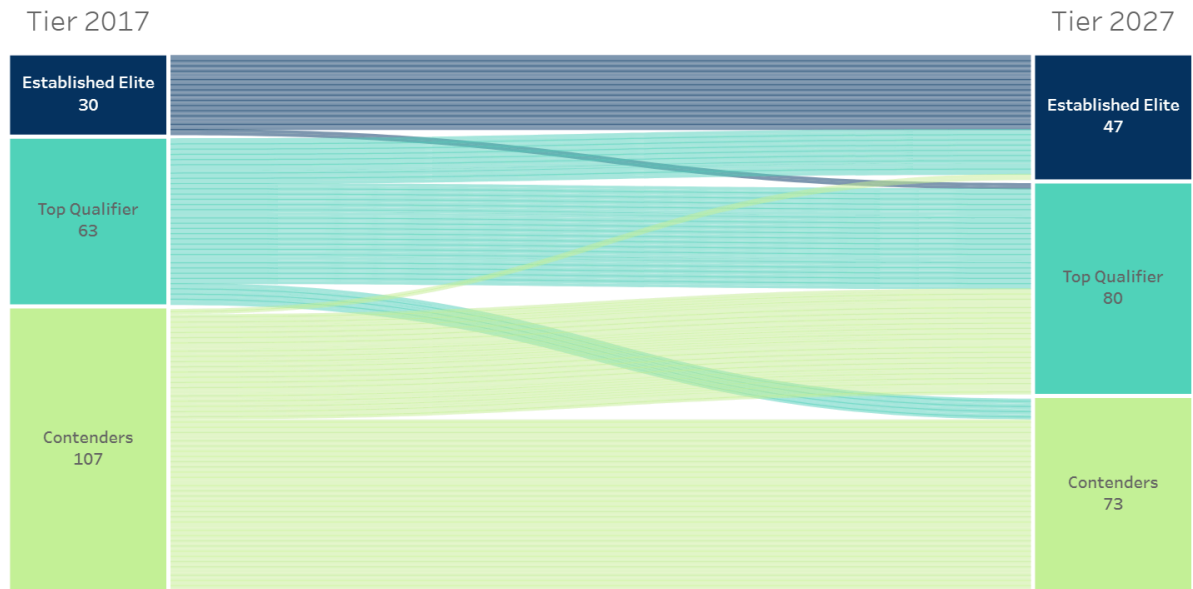


Figure 8. Sankey: Tier Mobility in Global Higher Education (2017-2027)

Figure 8 presents a Sankey diagram tracking movement between three performance tiers: Established Elite (top 50 by percentile rank), Top Qualifier (top 51–150), and Contenders (all others). These tiers divide the 200 highest-performing

universities as of 2017. This threshold was chosen because below the top 200, movement is overwhelmingly Contender-to-Contender and offers little insight into elite mobility specifically.

Findings show the global elite is largely stable. Of the 200 universities in this cohort, those reaching Established Elite status grew from 30 in 2017 to 47 by 2027, the Top Qualifiers growing to 80 from 63, while the Contenders tier shrank from 107 to 73. This indicates genuine, upward mobility rather than static prestige at the top. Movement is asymmetric: many Contenders and Top Qualifiers advanced, while few Elite universities declined; thus, entry into the global elite remains achievable but rare, and difficult to lose once reached.

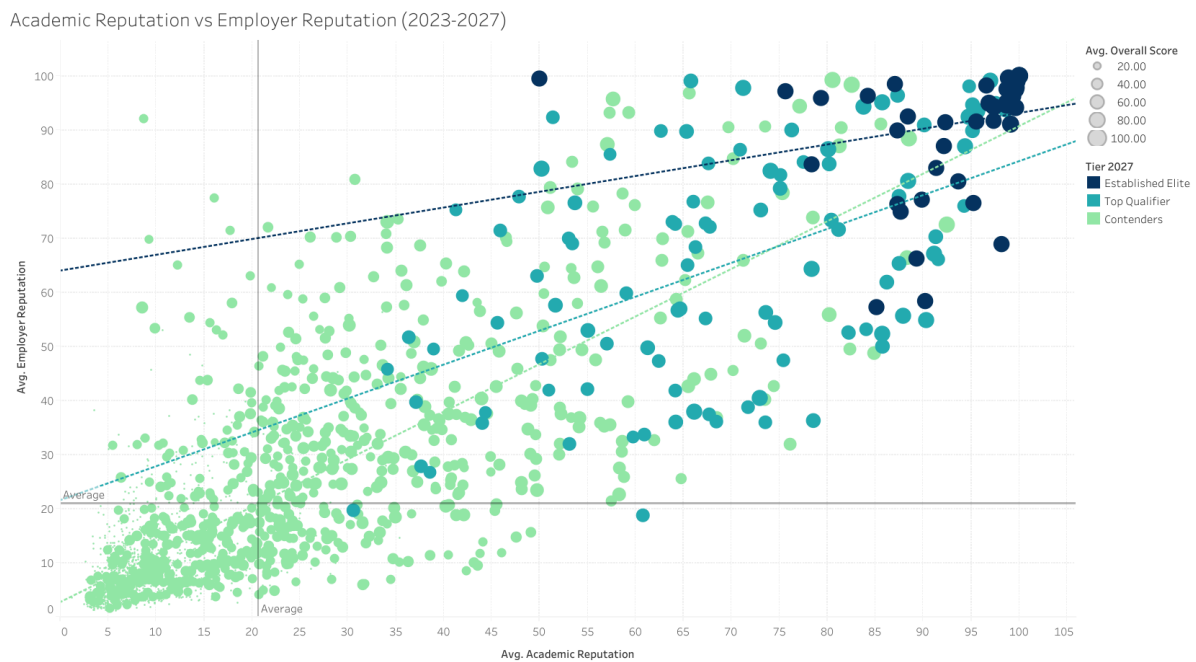


Figure 9. Scatter Plot: Academic Reputation vs Employer Reputation (2023-2027)

Figure 9 examines the same tiers in finer detail, plotting Academic against Employer Reputation for every university (2023–2027), coloured by Tier 2027 and sized by Overall Score. A clear positive relationship emerges: universities strong in one metric are generally strong in the other. Established Elite universities cluster tightly at the top; Contenders are most dispersed but concentrated at the lower end; Top Qualifiers sit above average with more variation. Research reputation and employer perception are strongly associated, more so at the elite end.

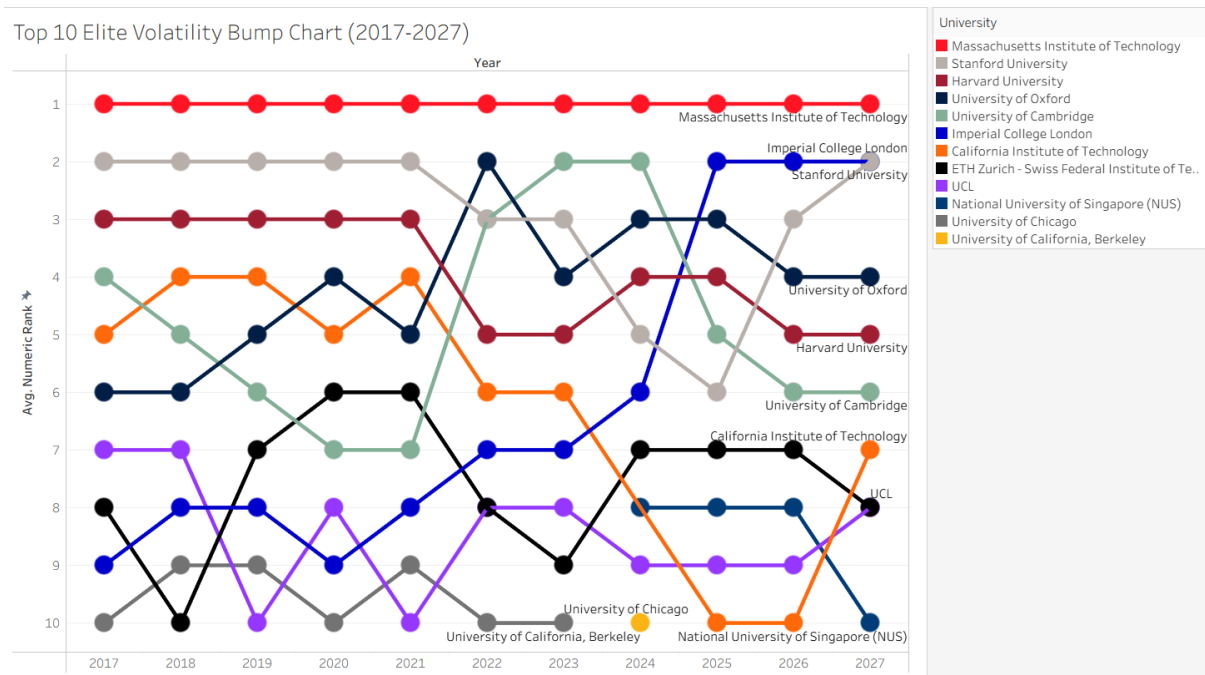


Figure 10. Bump Chart: Top 10 Elite Volatility Bump Chart (2017-2027)

Figure 10 tracks year-on-year rankings of the world's top ten universities (2017–2027). MIT held first place throughout; Stanford and Harvard remained leading institutions until 2021, with Stanford declining from 2022 before recovering in 2026, while Imperial College London showed the strongest upward trajectory. Though the top position never changed, the remaining elite competed closely, reflecting global higher education's dynamic nature.

Overall, global higher education remains concentrated within a small number of countries despite the rankings' expansion. Countries display distinct competitive strengths, while institutional mobility remains limited. These findings offer valuable business intelligence for universities, policymakers, and prospective students.

3. Selection of Data Mining Algorithm and Data Pre-processing

Time Series Forecasting was selected as the data mining algorithm, as the QS dataset consists of sequential numerical observations collected at equal yearly intervals between 2017 and 2027. This method identifies patterns in data recorded over time and uses them to project likely future values, which is well suited here because each country is tracked repeatedly across the same sequence of years, allowing genuine trend identification and extrapolation unlike a single cross-sectional snapshot, which cannot support forecasting.

This choice is supported directly by the visualisation analysis: the regional heatmap (Figure 2) and the English-medium country line chart (Figure 6). both revealed clear temporal patterns in average Percentile Rank across multiple years, indicating that forecasting could meaningfully extend this descriptive analysis by projecting likely future trends rather than only describing the past. The forecasting analysis uses Year as the time variable, Percentile Rank as the numerical measure of performance, and Country to segment the analysis, focusing on the English-medium countries examined in Figure 6.

Pre-processing was largely completed prior to visualisation and is detailed in Section 1: standardising column, country, and university names; converting ranking ranges (e.g., 601–610) to midpoints; and assigning 1401 to the open-ended 1401+ category. One additional anomaly was identified during the visualisation stage: Tableau's Data Interpreter was found to incorrectly parse university names containing commas, corrupting Level of Detail calculations used to derive Percentile Rank. This was resolved by disabling Data Interpreter and refreshing the data source, ensuring consistent aggregation for both visualisation and forecasting.

4. Data Mining

Building on the trends identified in Figure 6, time series regression was applied to the eleven English-medium countries to determine whether observed changes represented sustained long-term trends rather than short-term fluctuations. Figure 6 suggested that English-medium status alone does not determine competitiveness, as countries followed markedly different trajectories despite sharing a common language of instruction. Separate linear trend lines and forecasts (2017–2030) were generated in Tableau using Year as the independent variable and average Percentile Rank as the dependent variable. Tableau reports the coefficient of determination (R^2), from which the correlation coefficient (R) was calculated. Values closer to 1 indicate a stronger and more reliable trend.

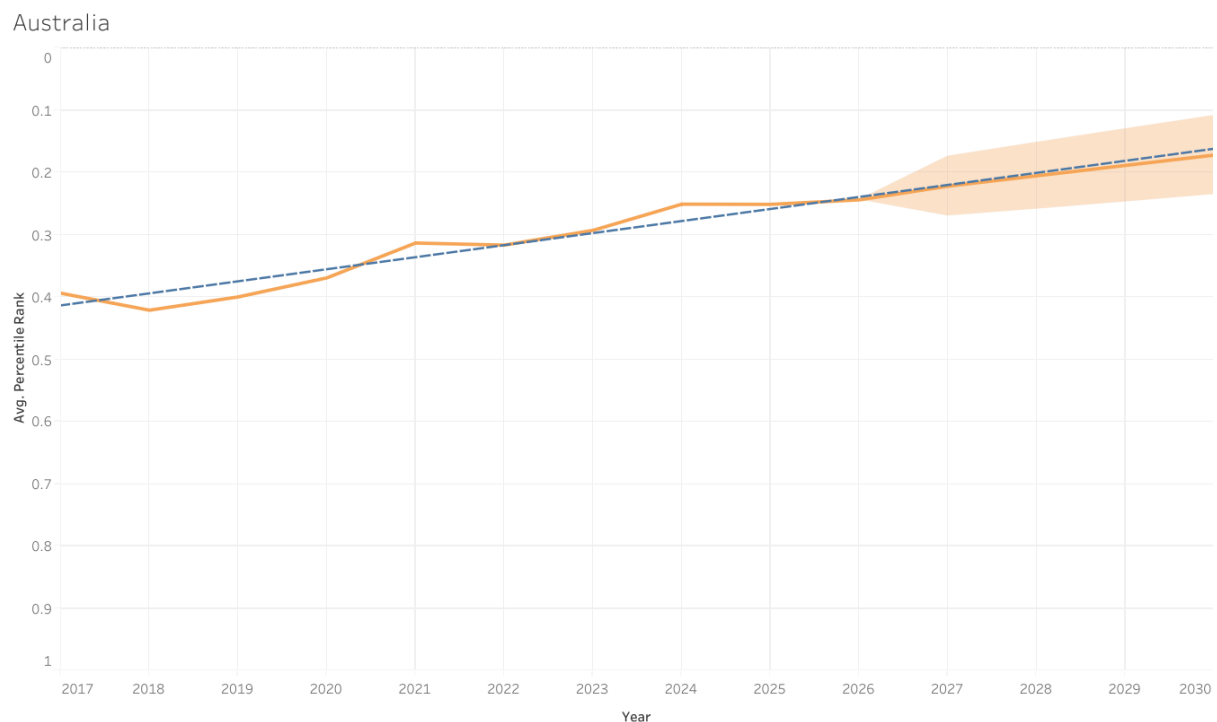


Figure 11. Australia ($R = 0.980$)

Australia shows a clear, consistent upward trend across the full study period, with the regression line closely tracking the observed data and minimal deviation. The excellent fit ($R = 0.980$) confirms this improvement is reliable rather than coincidental, and the forecast projects continued upward movement through 2030 with a narrow uncertainty band.

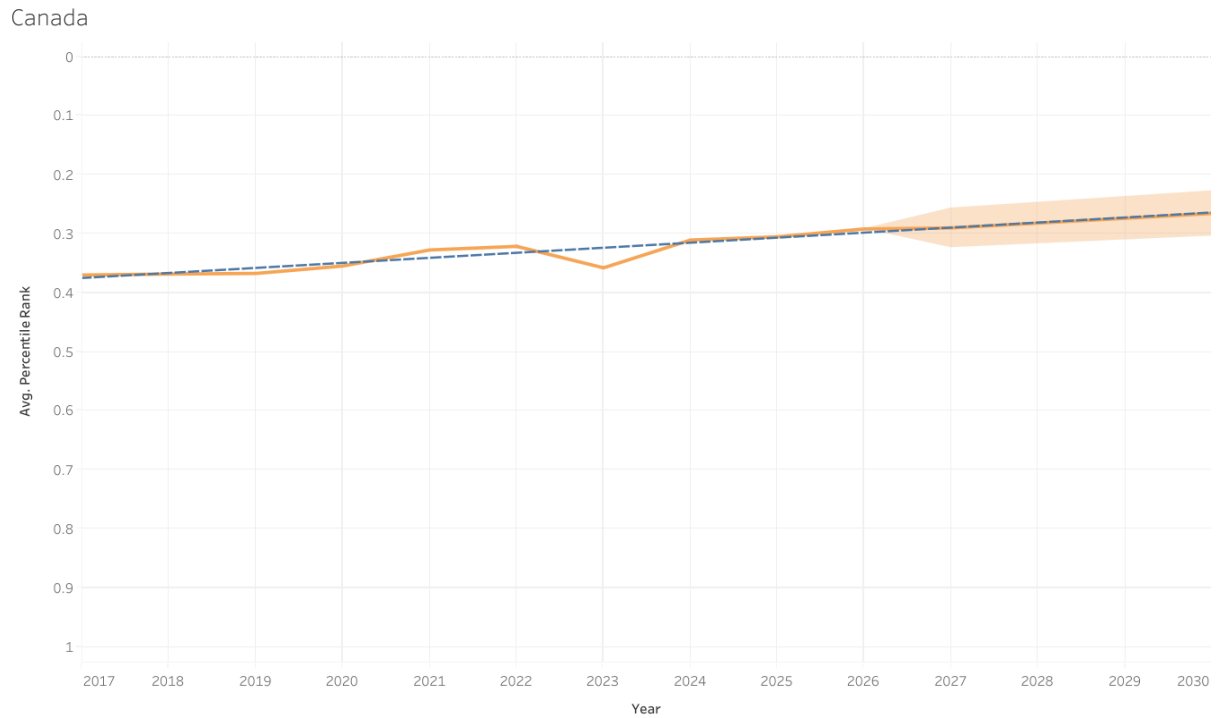


Figure 12. Canada ($R = 0.954$)

Canada demonstrates a steady upward trend throughout 2017–2027, with one notable dip in 2023 before recovering. Despite this, the regression line fits closely ($R = 0.954$), indicating a genuinely consistent long-term improvement. The forecast projects this trajectory continuing through 2030, with a moderately narrow uncertainty band reflecting the single-year deviation.

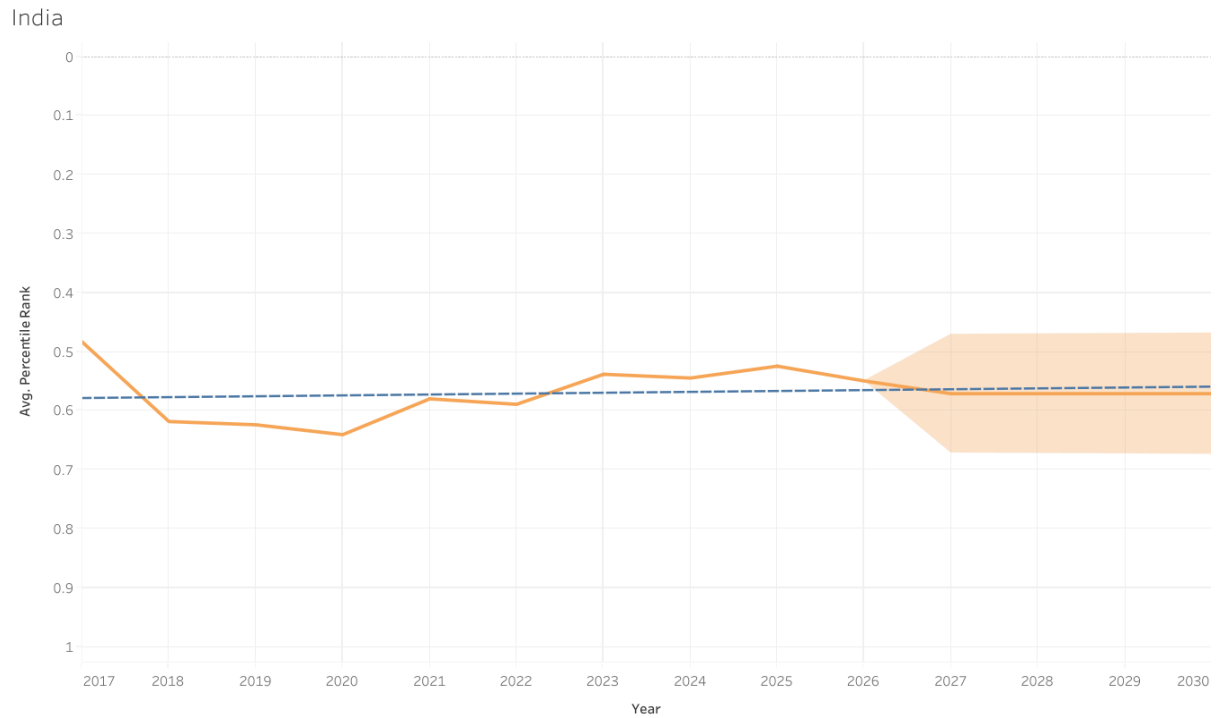


Figure 13. India ($R = 0.150$)

India exhibits considerable fluctuations across the study period without a clear long-term direction. The regression model produced an R-value of 0.150, indicating a very weak linear relationship. As a result, the forecast has limited predictive reliability and primarily reflects recent observations rather than a consistent long-term trend.

Ireland

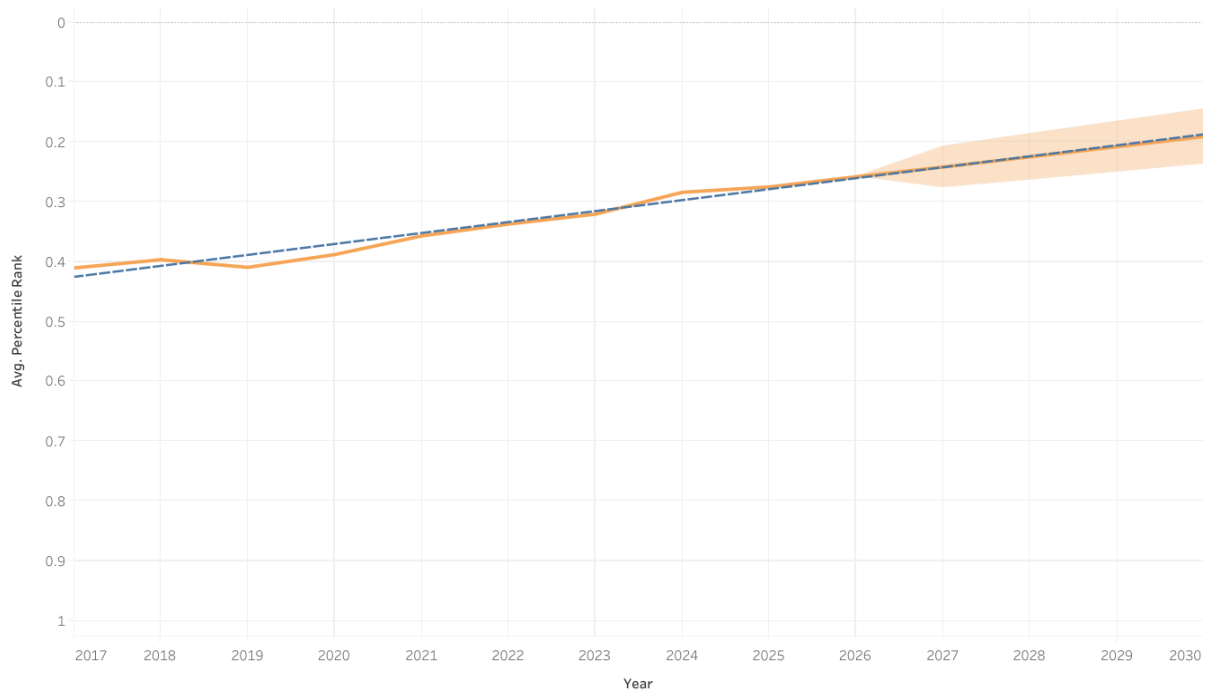


Figure 14. Ireland ($R = 0.991$)

Ireland produces the strongest fit of all eleven countries ($R = 0.991$), with the regression line almost indistinguishable from the observed trend line. A consistent, steady upward trajectory from 2017 onward leaves very little unexplained variation. The forecast projects continued improvement through 2030 with one of the narrowest uncertainty bands in this analysis.

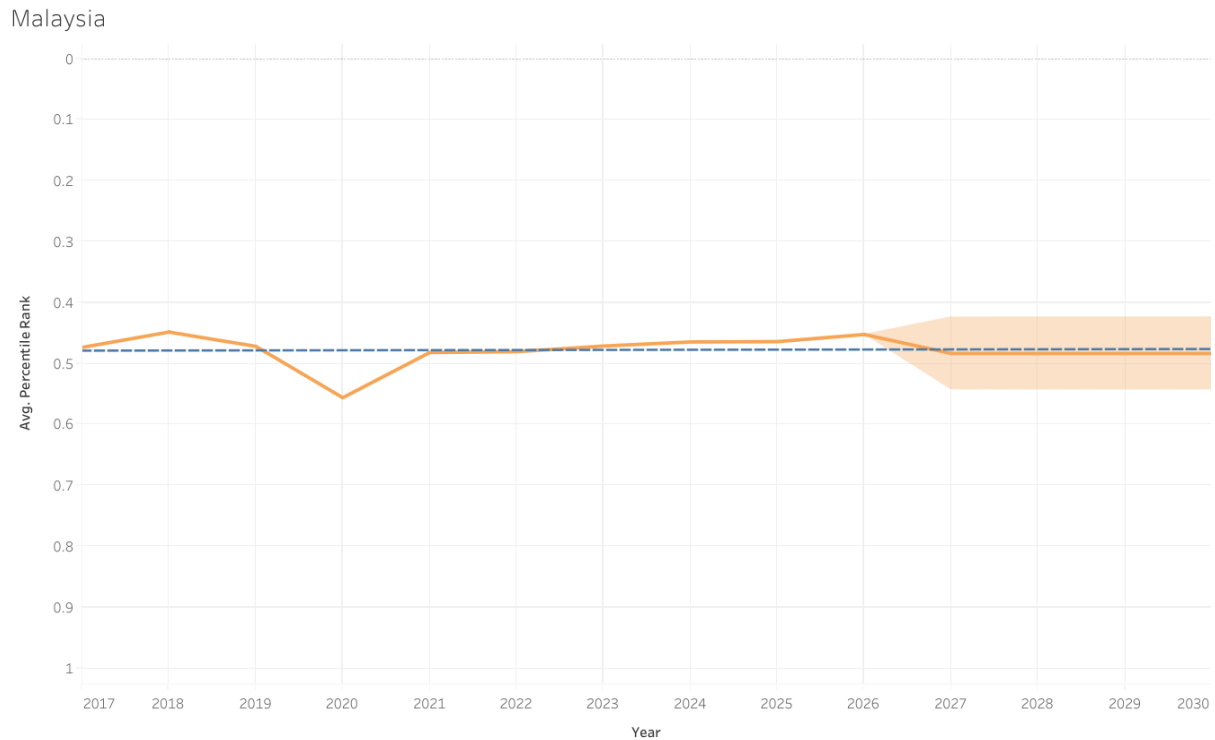


Figure 15. Malaysia ($R = 0.034$)

Malaysia's trend line is almost perfectly horizontal, reflecting a performance level that has remained essentially unchanged throughout the observation period despite short-term fluctuations. With an R-value of 0.034, the forecast predicts continued stability rather than substantial improvement, indicating virtually no linear relationship between year and percentile rank. Consequently, the forecast should be interpreted with caution.

NZ

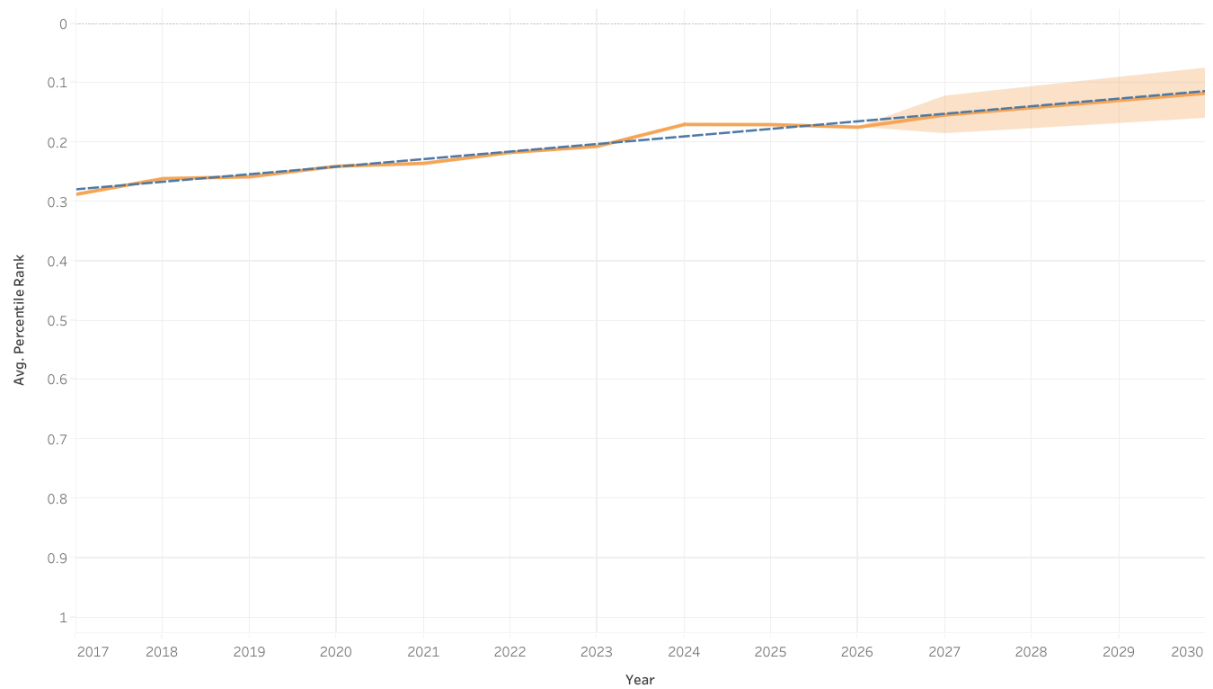


Figure 16. New Zealand ($R = 0.990$)

New Zealand shows one of the most consistent upward trends in the dataset ($R = 0.990$), with the observed line closely following the regression throughout, and almost no deviation. The forecast projects continued improvement through 2030 with a very narrow uncertainty band, making this one of the most reliable projections in this analysis.

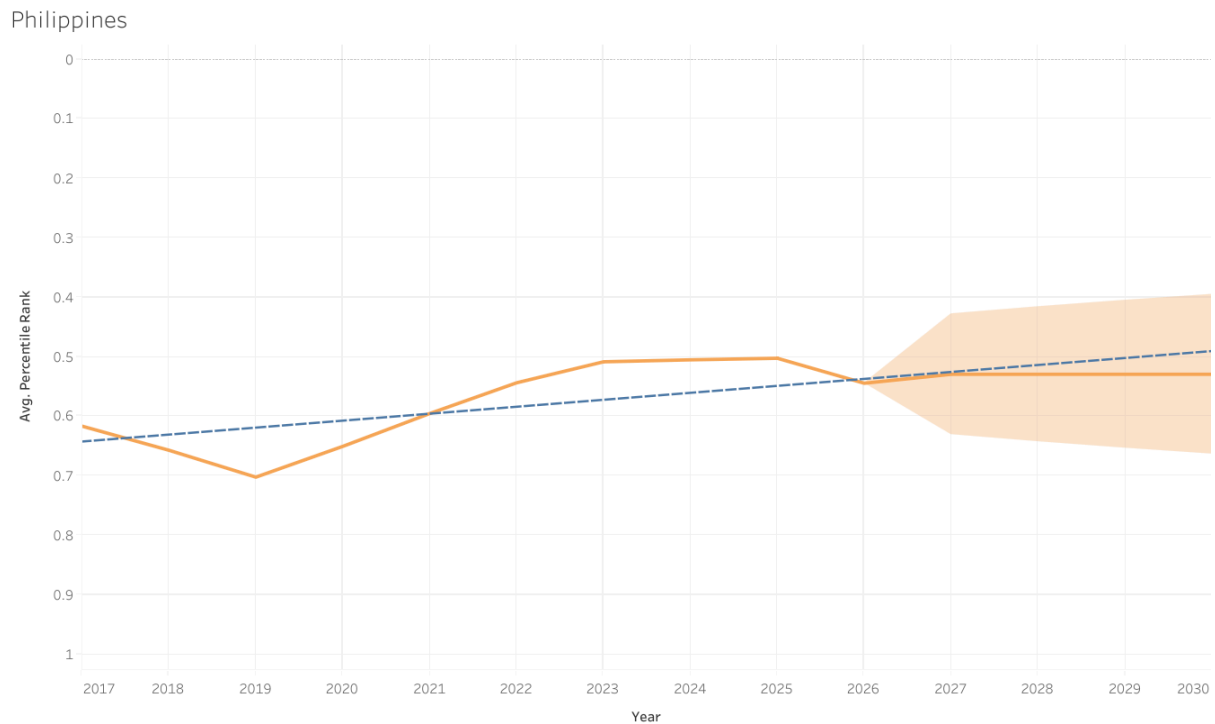


Figure 17. Philippines ($R = 0.754$)

The Philippines demonstrates noticeable improvement during the middle of the study period before stabilizing after 2025. The forecast suggests relatively stable performance through 2030. The regression model achieved an R- value of 0.754, indicating a reasonably strong trend despite some year-to-year variation.

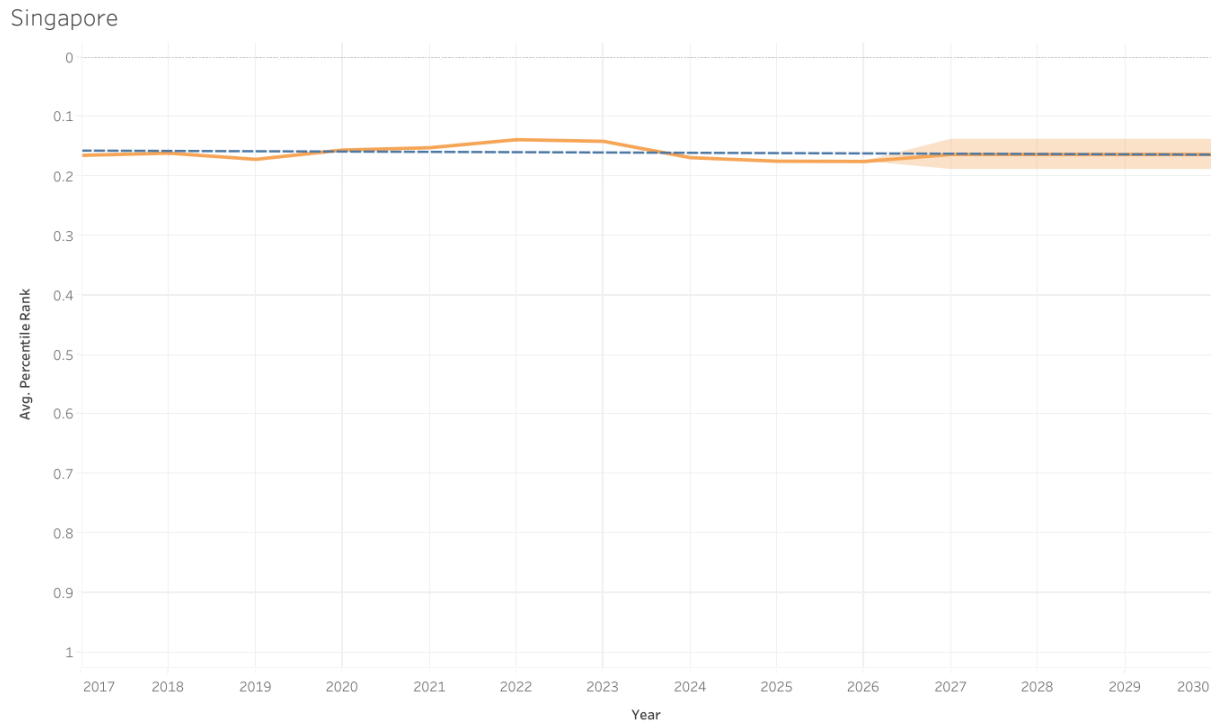


Figure 18. Singapore ($R = 0.195$)

Singapore's average Percentile Rank remains consistently strong and largely stable across the period, with the trend line nearly flat. The very low R-value ($R = 0.195$) and non-significant p-value (0.504) indicate no detectable linear trend. Singapore's performance is stable rather than improving or declining, and the forecast should be read as a projection of continued stability rather than a meaningful directional prediction.

South Africa

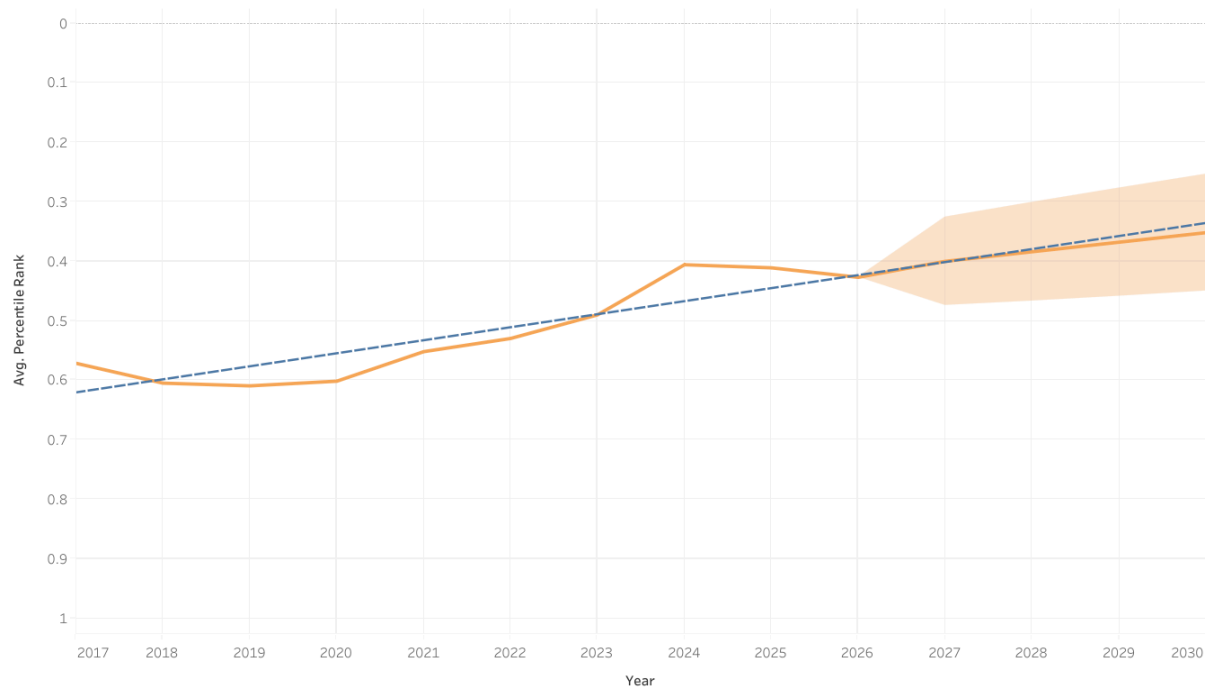


Figure 19. South Africa ($R = 0.950$)

South Africa demonstrates substantial improvement during the study period, particularly after 2022. The forecast indicates continued progress through 2030. The regression model achieved an R^2 value of 0.903 ($R = 0.950$), indicating an excellent fit and suggesting that the observed improvement follows a strong long-term trend.

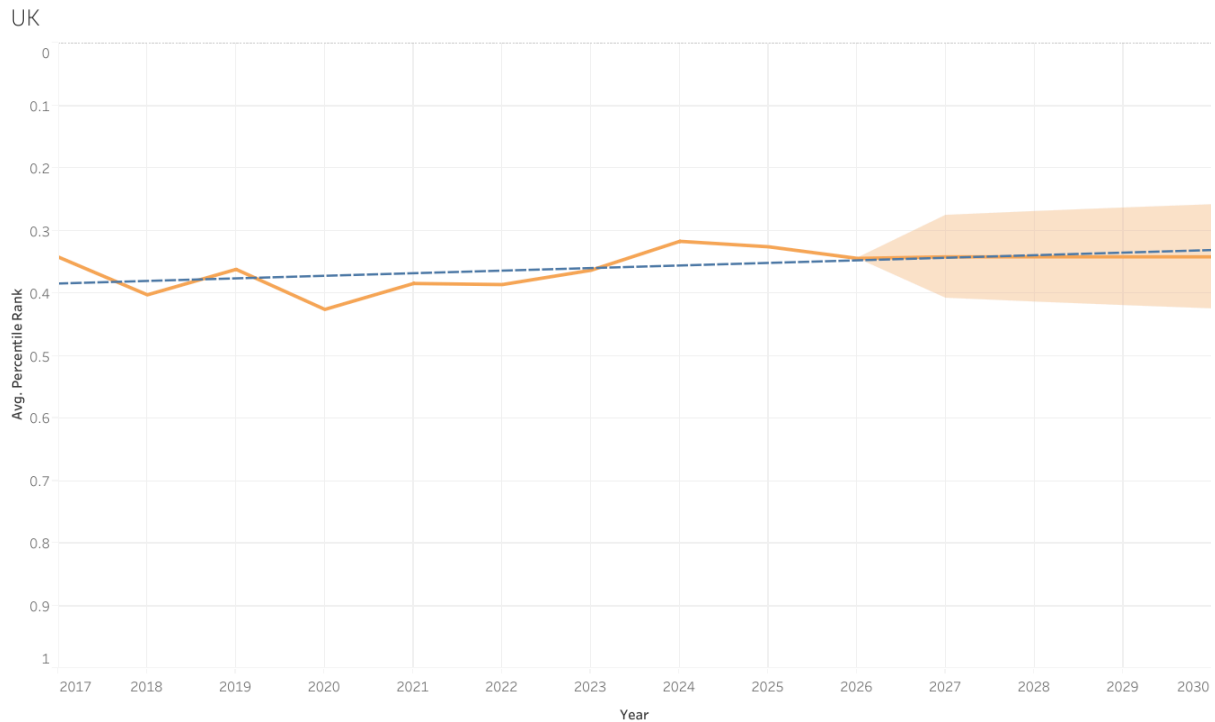


Figure 20. UK ($R = 0.559$)

The United Kingdom exhibits moderate fluctuations during the earlier years before stabilizing after 2023. Although the forecast predicts slight continued improvement, the regression model explains relatively little of the observed variation ($R = 0.559$). This suggests that UK university performance is influenced by greater year-to-year variation than countries with stronger linear trends.

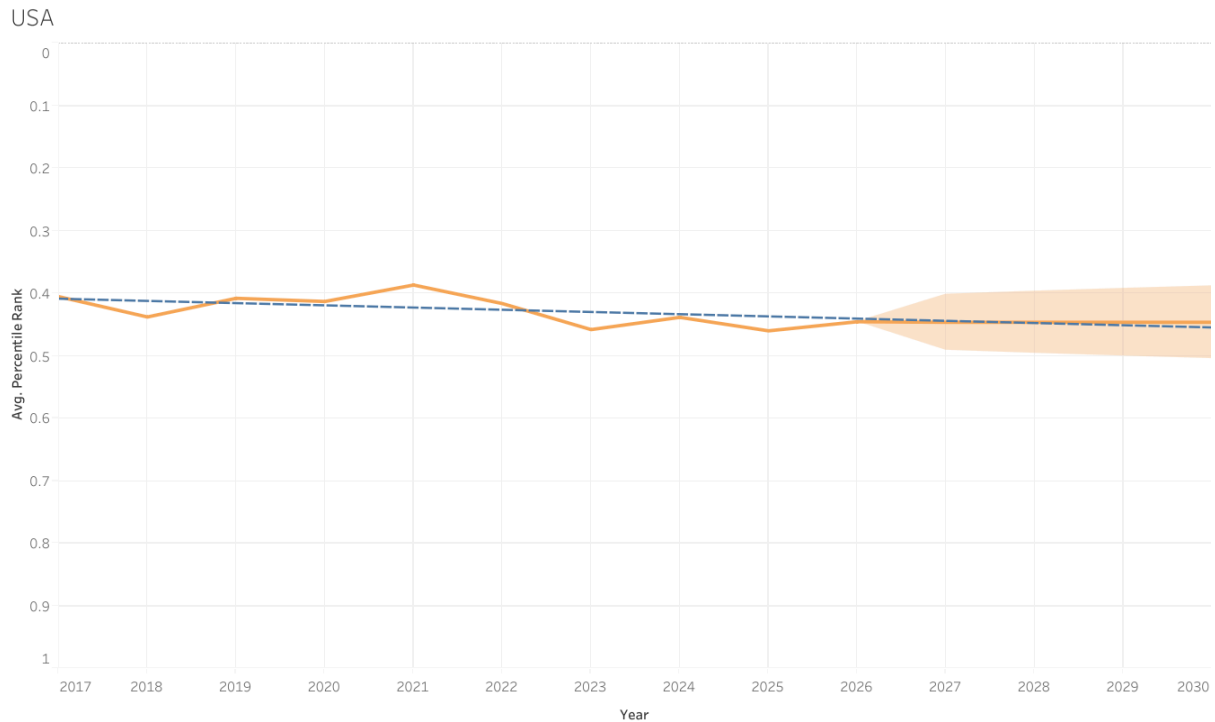


Figure 21. USA ($R = 0.663$)

The USA shows a moderate downward trend across the study period ($R = 0.663$), reflecting a gentle but consistent decline. As observed in Figure 6, this likely reflects an averaging effect: its large number of institutions lowers the national average percentile over time as lower-ranked institutions enter the pool.

Overall, the time series analysis reinforces the findings from Section 2. Australia, Canada, Ireland, New Zealand, and South Africa exhibit highly consistent long-term improvement ($R > 0.95$), whereas the United Kingdom, United States, and Philippines display moderate trends with greater annual variation. India, Malaysia, and Singapore reflect more stability than directional change. These forecasts suggest that the competitive landscape will remain broadly stable through 2030, although projections should be interpreted cautiously given the relatively short eleven-year time series.

5. Data Ethics

Business intelligence raises ethical, legal, and professional questions even when, as here, the dataset contains no personal data. Ethical issues still arise from how ranking data is interpreted and presented.

University rankings shape real decisions made by students, parents, universities, and governments, so misleading charts or careless interpretation could lead people to draw the wrong conclusions about a country or institution's quality. Following the five principles of ethical data use, which consists of being fair, respectful, and beneficial to those affected, this analysis aimed to present findings honestly, including limitations, rather than only the most flattering patterns.

Legally, the dataset was sourced from a public Kaggle repository, and users of such data must respect copyright and licensing terms. While QS ranking data is not personal data and falls largely outside UK GDPR, it should still be stored and shared responsibly. Professionally, the BCS Code of Conduct calls for public interest and professional integrity, which means cleaning decisions should be documented so the work can be checked or repeated by others.

Analysts should avoid letting personal bias shape their interpretation and should clearly flag limitations, such as missing sustainability data in certain years and changes to QS's own methodology over time. Overall, being open about how the data was collected, cleaned, and interpreted is what makes business intelligence findings trustworthy and genuinely useful for decision-making.

6. Conclusion

This project analyzed eleven years of QS World University Rankings data to identify long-term patterns in global higher education performance using Tableau visualizations and time series forecasting. Normalizing rankings through Percentile Rank enabled fair comparison despite the substantial expansion of the QS rankings between 2017 and 2027.

The visualization analysis revealed that global higher education remains concentrated within a relatively small number of countries. Western Europe, North America, and Oceania consistently achieved the strongest overall performance, while countries demonstrated different competitive strengths across the individual QS indicators rather than excelling uniformly. Although movement between performance tiers occurs, the world's leading universities remain relatively stable. The comparison of English-medium countries further demonstrated that language alone does not determine competitiveness; national investment, institutional quality, and the composition of higher education systems appear to play more significant roles.

Time series forecasting reinforced these findings by showing that countries such as Ireland, New Zealand, Australia, Canada, and South Africa exhibited highly consistent long-term trends with strong regression fit (high R-values), making their forecasts comparatively reliable. Conversely, India, Malaysia, and Singapore showed weak linear relationships with time, indicating stable or fluctuating performance rather than sustained directional change. Overall, the forecasting models suggest that the competitive landscape is unlikely to change dramatically in the short term.

The findings provide valuable business intelligence for multiple stakeholders. Universities can benchmark their performance against comparable institutions and identify areas requiring strategic investment. Policymakers can evaluate the effectiveness of national higher education systems and prioritize funding to improve global competitiveness. Prospective students can use these insights to compare countries and institutions using long-term performance trends rather than single-year rankings, supporting more informed decisions about international study.

For all three groups, the forecasts provide an evidence-based basis for near-term strategic planning, while the data quality findings serve as a reminder that ranking data should be interpreted with awareness of its methodological constraints rather than treated as a neutral, objective measure of quality.